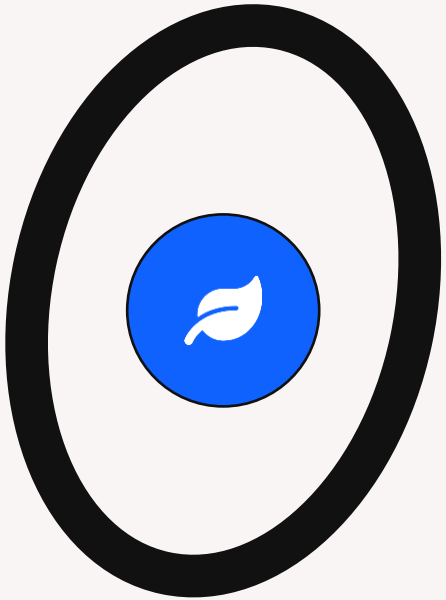
IBM Z DATATHON 2026 · TRACK 1

Team Infinityhacks

NourishNet AI

Real-Time Food Rescue & Intelligent Redistribution Engine

Challenge: Real-Time AI for Critical Decisions. Sub -15 ms autonomous dispatch that rescues surplus food before it spoils - explainable, auditable, built for IBM LinuxONE.



2.4 ms	66 / 66	100 / 100	Surplus → Safe → Served
Decision latency	Tests passed	Track 1 rubric	

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Problem Statement: The Perishability Trap

1.3 Billion Tons Wasted

One-third of all food is discarded while 828 million people face acute hunger; waste drives 8–10% of global greenhouse gases.

2–4 Hour Spoilage Window

Cooked meals and dairy enter the 5–60 °C Danger Zone at once; manual calls and batch pickups arrive too late.

Donor Fear & Liability Trap

Without cold-chain proof, hotels and caterers dump surplus rather than risk civil liability.

Why LLM chatbots and static routing fail

- Hallucinations and non-deterministic math
- Multi-second API latency
- Ignores temperature, humidity and traffic
- Ignores NGO capacity quotas

We need a sub-second, mathematically rigorous decision engine.

Proposed Solution: 4-Engine AI Pipeline

1 BIOPHYSICS

Arrhenius Spoilage Predictor

Predicts microbial decay from temperature, humidity and packaging.

Dynamic safe seconds

2 OPTIMIZATION

Kuhn-Munkres Matcher

Globally optimal donor-to-NGO assignment in $O(V^3)$.

Optimal dispatch

3 TRUST & EQUITY

TreeSHAP + Gini Audit

Explains every dispatch and audits fairness across NGOs.

Transparent attribution

4 SECURITY

SHA-256 Custody Vault

Tamper-evident chain from donor intake to NGO handoff.

Liability certificate

Surplus Alert → **Arrhenius Model** → **Hungarian Match** → **SHAP Audit** → **Crypto Vault**

End-to-end pipeline latency: 2.4 ms

Engine 1: Dynamic Spoilage Prediction

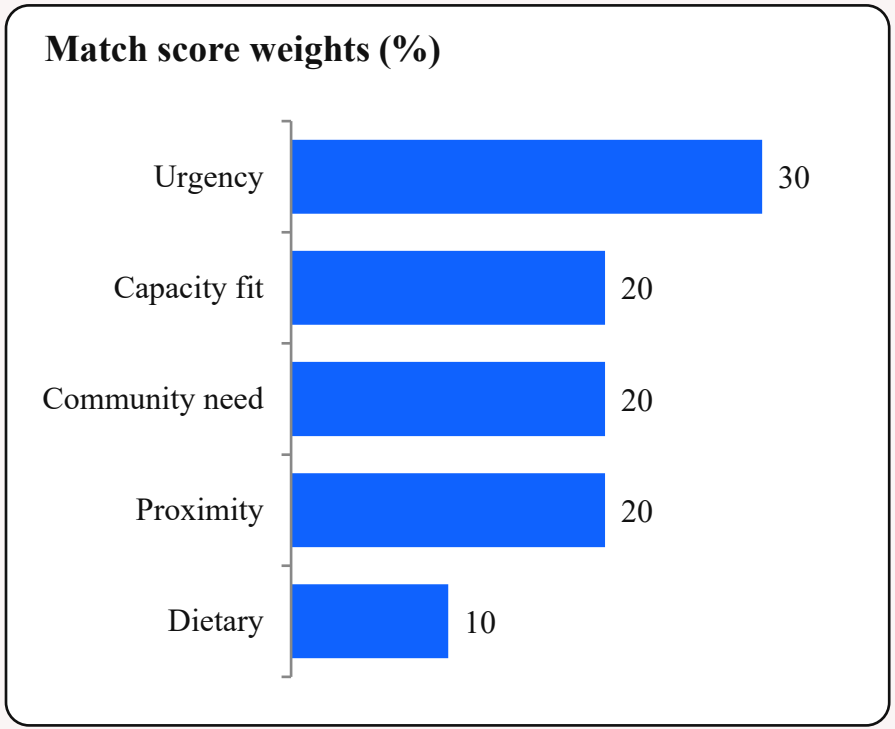
MODIFIED ARRHENIUS MODEL

t_safe = t_0 x exp(-k_decay x (T_ambient – T_base) / 10) x α_humidity x ß_packaging

- t_0: baseline safe hours at base temperature
- k: thermal sensitivity (0.25 leafy greens, 0.18 cooked meals)
- α_humidity = 1 − 0.003 × max(0, H − 40)
- β_packaging: sealed 1.0×, open 0.7×, vacuum 1.5×, frozen 3.0×

>60 min	**CRITICAL** Immediate override; nearest shelter within 5 km
60–180 min	**URGENT** Priority match with community kitchens
180–480 min	**NORMAL** Batch optimisation across food banks
≥ 480 min	**STABLE** Canned and dry goods; multi-drop routes

Engine 2: Kuhn-Munkres Optimal Matching



Score(D, N) = w_1 S_urgency + w_2 S_capacity + w_3 S_need + w_4 S_proximity + w_5 S_dietary

Hard constraint: Safe transit buffer

ETA + 15 min > remaining safe time ⇒ REJECT

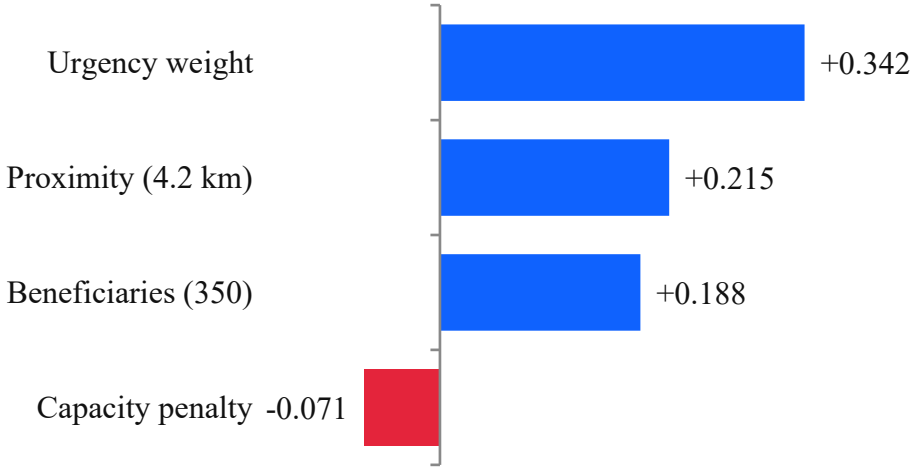
Hard constraint: Capacity ceiling

Donation > 2× NGO storage capacity ⇒ REJECT

Zero-API dependency: runs on OpenStreetMap with on-chip geodesics (<3 ms).

Engine 3: Explainable and Fair Decisions

Why NGO-A? Local TreeSHAP attribution



Shown live as an interactive attribution chart in the dispatcher UI.

0.284

Measured live Gini index (target ≤ 0.45)

Excellent equity

Fairness guardrail

NGOs below 15% of mean allocation get boost multipliers in the solver, ending “food aid deserts”.

Engine 4: Cryptographic Chain-of-Custody

Hash_n = SHA256(CanonicalJSON(Transaction_n) || Hash_{n-1})

Block 0

Genesis

Anchor block

Block 1

Donor intake

Category, quantity, temp

Block 2

Cold-chain transit

Ambient and drop-off temp

Block 3

NGO handoff

Timestamp, signatures

Tamper detection

Changing a single gram or timestamp invalidates every downstream block instantly.

Good Samaritan Safe Haven Certificate

A legally admissible PDF proving cold-chain continuity, unlocking corporate food donations.

Built for IBM LinuxONE (s390x)

Hardware crypto (CPACF)

On-core SHA-256 and AES at wire-speed with near-zero CPU overhead.

99.99999% reliability

Seven-nines uptime keeps dispatch live through storms and grid failures.

75% less energy

Consolidating microservices on one LinuxONE partition cuts carbon.

Production topology

- Core API: FastAPI + Uvicorn async workers
- AI math: NumPy and SciPy matrix ops
- Storage: SQLite with WAL
- UI: React 18 SPA + Vite

Mission Command Center UI

- One-click single or batch AI dispatch
- Live shelf-life as donors adjust sliders
- Urgency-coded pins on OpenStreetMap
- Instant SHA-256 certificate viewer

Impact: Quantified ESG Value

160 kg

Surplus food rescued

400

Nutritious meals served

400 kg

CO_2e emissions prevented

16,000 L

Freshwater conserved

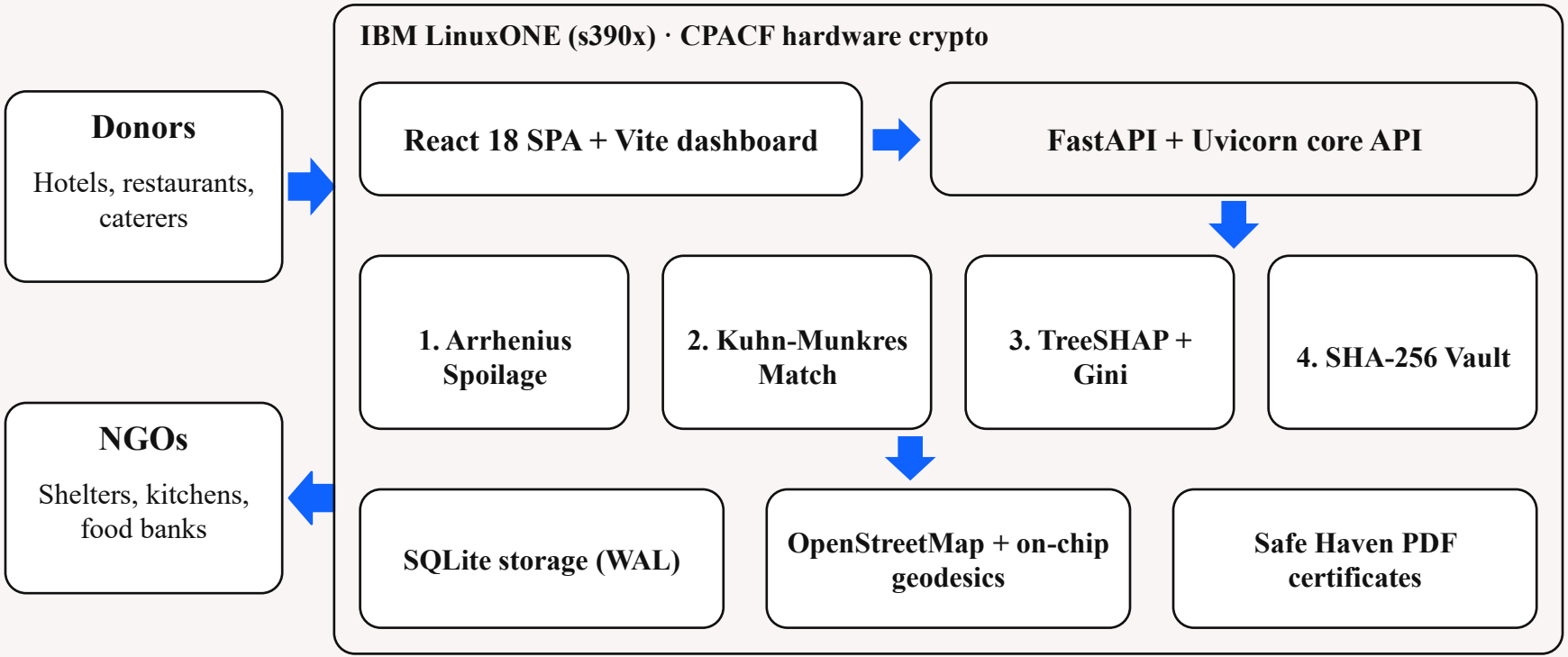
Corporate ESG reporting

Donors receive certified Scope 3 emissions data for GRI, CSRD and SEC sustainability filings, backed by the SHA-256 audit trail.

UN SDG alignment

- **SDG 2 · Zero Hunger**
- **SDG 11 · Cities**
- **SDG 12 · Responsible Use**
- **SDG 13 · Climate Action**

System Architecture Diagram



Roadmap: Prototype to Enterprise Scale

Phase 1 · Complete

- 4 AI engines
- Sub-15 ms decision latency
- OpenStreetMap live map
- SHA-256 custody vault
- 66/66 master test suite

Phase 2 · Next 60 days

- BLE / IoT temperature loggers
- Automated real-time cold-chain logging
- Live telemetry into Arrhenius engine

Phase 3 · Enterprise scale

- Native IBM LinuxONE s390x deployment
- CPACF crypto co-processing
- Telum II on-chip AI inference

Goal: automated cold-chain proof and mission-critical scale on IBM Z

Technology Stack

Frontend

- React 18 SPA
- Vite bundler
- OpenStreetMap tiles

Backend

- FastAPI
- Uvicorn async workers
- REST dispatch API

AI & Math

- NumPy and SciPy
- Kuhn-Munkres solver
- TreeSHAP, Gini audit

Data

- SQLite with WAL
- Append-only custody ledger
- Canonical JSON records

Security

- SHA-256 hash chain
- Safe Haven PDF certificates
- Hardware crypto (CPACF)

Infrastructure

- IBM LinuxONE (s390x)
- 7-nines availability
- Telum II inference (planned)